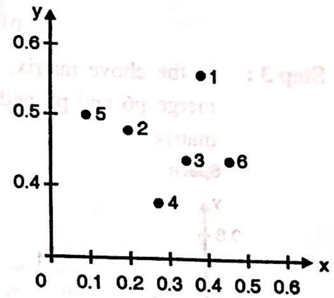
**Ex. 9.5.1 :** Assume that the database D is given by the table below. Follow single link technique to find clusters in D. Use Euclidean distance measure.

| | | X | Y |
|---|---|---|---|
| | p1 | 0.40 | 0.53 |
| | p2 | 0.22 | 0.38 |
| D = | p3 | 0.35 | 0.32 |
| | p4 | 0.26 | 0.19 |
| | p5 | 0.08 | 0.41 |
| | p6 | 0.45 | 0.30 |

**Soln. :**

**Step 1 :** Plot the objects in $n$-dimensional space (where $n$ is the number of attributes). In our case we have 2 attributes x and y, so we plot the objects p1, p2, … p6 in 2-dimensional space :




**Step 2 :** Calculate the distance from each object (point) to all other points, using Euclidean distance measure, and place the numbers in a distance matrix.

The formula for Euclidean distance between two points i and j is :

$$d(i, j) = \sqrt{|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j1}|^2 + \ldots + |x_{ip} - x_{jp}|^2}$$

where $x_{i1}$ is the value of attribute 1 for i and $x_{j1}$ is the value of attribute 1 for j and so on, as many attributes we have … shown up to p i.e. $x_{ip}$ in the formula.

In our case, we only have 2 attributes. So, the Euclidean distance between our points p1 and p2, which have attributes x and y would be calculated as follows:

$$\begin{aligned} d(p1, p2) &= \sqrt{|xp_1 - xp_1|^2 + |yp_1 - yp_2|^2} \\ &= \sqrt{|0.40 - 0.22|^2 + |0.53 - 0.38|^2} \\ &= \sqrt{|0.18|^2 + |0.15|^2} \\ &= \sqrt{0.0324 + 0.0225} \\ &= \sqrt{0.0549} = 0.2343 \end{aligned}$$

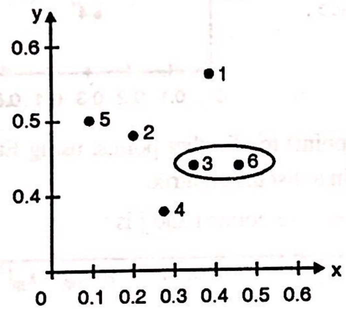
Analogically, we calculate the distance to the remaining points and we will receive the following values :

**Distance matrix :**

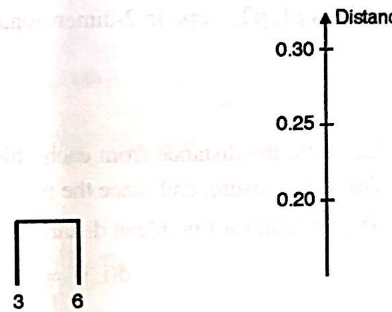
| | p1 | p2 | p3 | p4 | p5 | p6 |
|---|---|---|---|---|---|---|
| p1 | 0 | | | | | |
| p2 | 0.24 | 0 | | | | |
| p3 | 0.22 | 0.15 | 0 | | | |
| p4 | 0.37 | 0.20 | 0.15 | 0 | | |
| p5 | 0.34 | 0.14 | 0.28 | 0.29 | 0 | |
| p6 | 0.23 | 0.25 | **0.11** | 0.22 | 0.39 | 0 |

**Step 3 :** In the above matrix, p6 and p3 are two clusters with shortest distance 0.11, so merge p6 and p3 and make single cluster (p3,p6). Now re-compute the distance matrix.

Space




Dendrogram




**Distance matrix :**

| | p1 | p2 | (p3, p6) | p4 | p5 |
|---|---|---|---|---|---|
| p1 | 0 | | | | |
| p2 | 0.24 | 0 | | | |
| **(p3, p6)** | **0.22** | **0.15** | 0 | | |
| p4 | 0.37 | 0.20 | **0.15** | 0 | |
| p5 | 0.34 | 0.14 | **0.28** | 0.29 | 0 |

To calculate the distance of p1 from (p3,p6) :

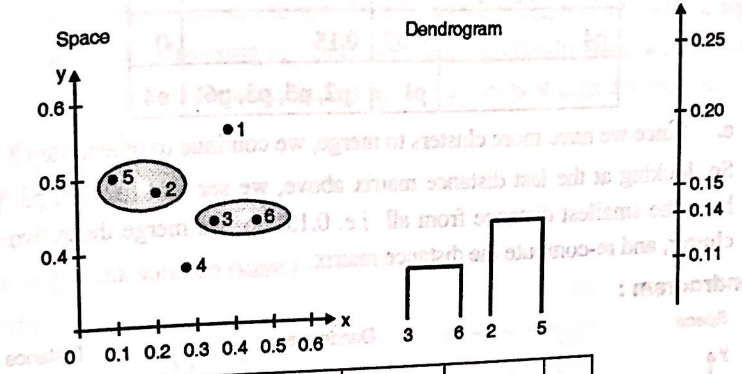
dist( (p3, p6), p1) = MIN ( dist(p3, p1) , dist(p6, p1) )

= MIN (0.22, 0.23) //from original matrix

= 0.22

**Step 4 :** Repeat Step 3 until one single cluster is formed i.e. merge all the clusters.

a. Now p2 and p5 have the smallest distance from above matrix, so merge p2 and p5 to get cluster(p2,p5).




| | | | | |
|---|---|---|---|---|
| p1 | 0 | | | |
| **(p2, p5)** | **0.24** | 0 | | |
| (p3, p6) | 0.22 | **0.15** | 0 | |
| p4 | 0.37 | **0.20** | 0.15 | 0 |
| | p1 | **(p2, p5)** | (p3, p6) | p4 |

The distance between (p3, p6) and (p2, p5) is calculated as given below :

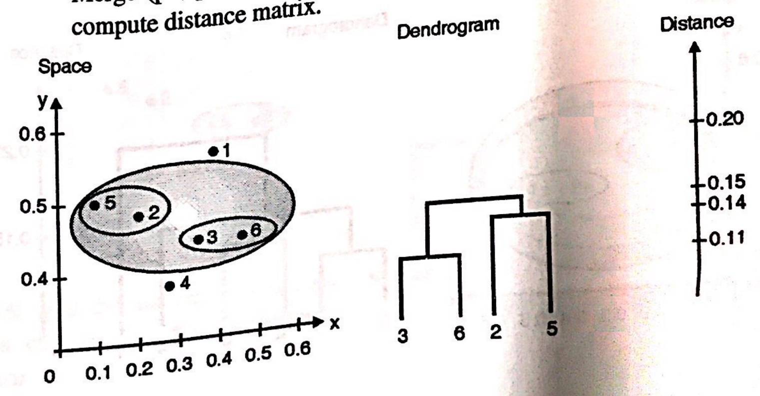
dist( (p3, p6), (p2, p5) ) = MIN ( dist(p3, p2) , dist(p6, p2), dist(p3, p5), dist(p6, p5))

= MIN (0.15, 0.25, 0.28, 0.39) //from original matrix

= 0.15

b. Repeat Step 3.

Merge (p2, p5) and (p3, p6) as having minimum distance i.e.0.15 and again compute distance matrix.



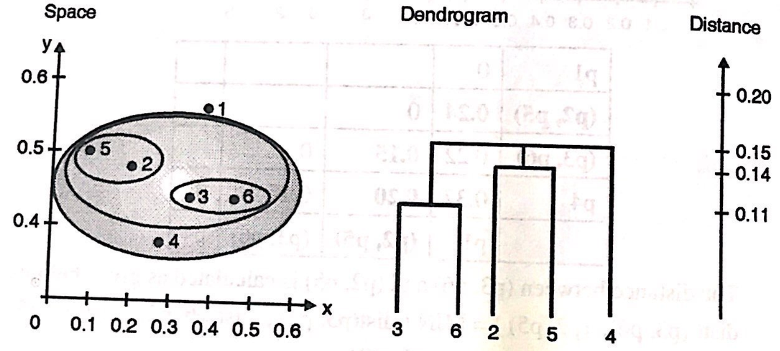
**Distance matrix :**

| | | | |
|---|---|---|---|
| p1 | 0 | | |
| (p2, p5, p3, p6) | **0.22** | 0 | |
| p4 | 0.37 | **0.15** | 0 |
| | p1 | (p2, p5, p3, p6) | p4 |

c. Since we have more clusters to merge, we continue to repeat Step 3.

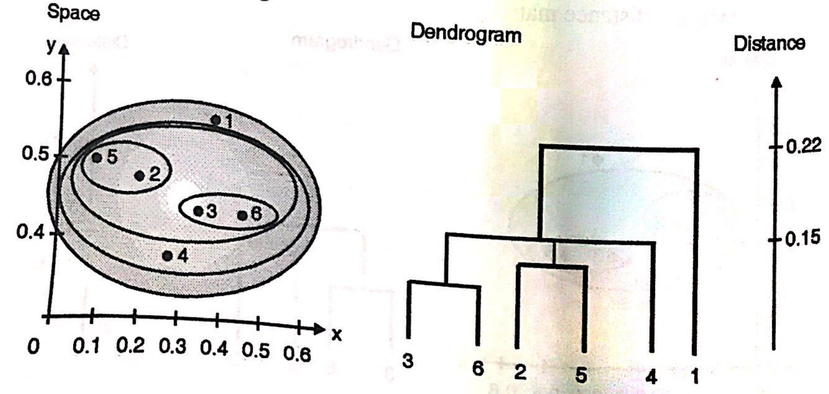
So, looking at the last distance matrix above, we see that (p2, p5, p3, p6) and p4 have the smallest distance from all i.e. 0.15. So, we merge those two in a single cluster, and re-compute the distance matrix.

**Space dendrogram :**




d. Since we have more clusters to merge, we continue to repeat Step 3.

So, looking at the last distance matrix above, we see that (p2, p5, p3, p6, p4) and p1 have the smallest distance - 0.22 (the only one left). So, we merge those two in a single cluster. There is no need to re-compute the distance matrix, as there are no more clusters to merge.



### Stopping condition :

We indicated that "each object is placed in a separate cluster, and at each step we merge the closest pair of clusters, *until certain termination conditions are satisfied*".

To stop clustering either user has to specify the number of clusters he wants or algorithm has to make decision to stop clustering at which level. Through dendrogram, we can notice the merging of clusters at various distances. If merging of clusters is at high distance then we can stop clustering at that level.

### Complete link :

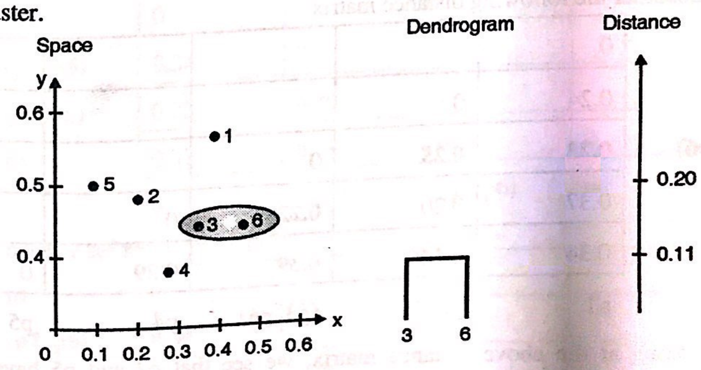
**Step 1 and step 2 :**

Refer the single link solution (same)

**Distance matrix :**

| | p1 | p2 | p3 | p4 | p5 | p6 |
|---|---|---|---|---|---|---|
| p1 | 0 | | | | | |
| p2 | 0.24 | 0 | | | | |
| p3 | 0.22 | 0.15 | 0 | | | |
| p4 | 0.37 | 0.20 | 0.15 | 0 | | |
| p5 | 0.34 | 0.14 | 0.28 | 0.29 | 0 | |
| p6 | 0.23 | 0.25 | 0.11 | 0.22 | 0.39 | 0 |

**Step 3 :** Identify the two clusters with the shortest distance in the matrix, and merge them together. Re-compute the distance matrix, as those two clusters are now in a single cluster.




By looking at the distance matrix above, we see that p3 and p6 have the *smallest* distance from all i.e. 0.11 So, we merge those two in a single cluster and *re-compute* the distance matrix.

*dist*( (p3, p6), p1) = MAX ( *dist*(p3, p1) , *dist*(p6, p1) )
= MAX (0.22, 0.23) //from original matrix
= 0.23

*dist*( (p3, p6), p2) = MAX ( *dist*(p3, p2) , *dist*(p6, p2) )
= MAX (0.15, 0.25) //from original matrix
= 0.25

*dist*( (p3, p6), p4) = MAX ( *dist*(p3, p4) , *dist*(p6, p4) )
= MAX (0.15, 0.22) //from original matrix
= 0.22

*dist*( (p3, p6), p5) = MAX ( *dist*(p3, p5) , *dist*(p6, p5) )
= MAX (0.28, 0.39) //from original matrix
= 0.39

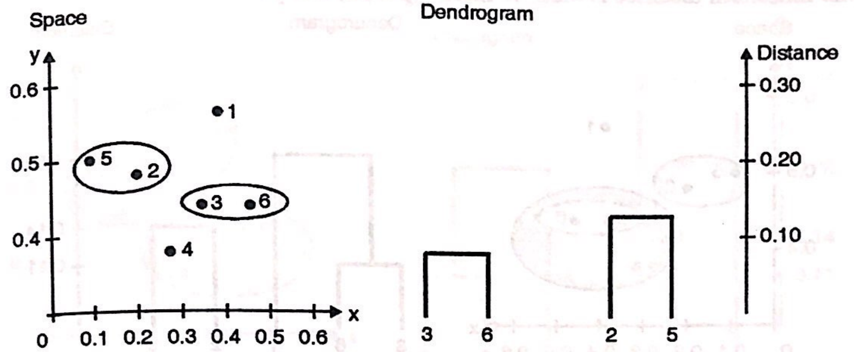
**New distance matrix :**

| | p1 | p2 | (p3, p6) | p4 | p5 |
|---|---|---|---|---|---|
| p1 | 0 | | | | |
| p2 | 0.24 | 0 | | | |
| (p3, p6) | **0.23** | **0.25** | 0 | | |
| p4 | 0.37 | 0.20 | **0.22** | 0 | |
| p5 | 0.34 | 0.14 | **0.39** | 0.29 | 0 |

**Step 4 :** Consider the following distance matrix

| | p1 | p2 | (p3, p6) | p4 | p5 |
|---|---|---|---|---|---|
| p1 | 0 | | | | |
| p2 | 0.24 | 0 | | | |
| (p3, p6) | **0.23** | **0.25** | 0 | | |
| p4 | 0.37 | 0.20 | **0.22** | 0 | |
| p5 | 0.34 | 0.14 | **0.39** | 0.29 | 0 |

So, looking at the above distance matrix, we see that p2 and p5 have the smallest distance from all – 0.14. So, we merge those two in a single cluster, and re-compute the distance matrix using the following calculations.




*dist*( (p2, p5), p1) = MAX ( *dist*(p2, p1) , *dist*(p5, p1) )
= MAX (0.24, 0.34) //from original matrix
= 0.34

*dist*( (p2, p5), (p3,p6)) = MAX ( *dist*(p2, p3) , *dist*(p2, p6), *dist*(p5, p3), *dist*(p5, p6) )
= MAX (0.15, 0.25, 0.28, 0.39)
//from original matrix
= 0.39

*dist*( (p2, p5), p4) = MAX ( *dist*(p2, p4) , *dist*(p5, p4) )
= MAX (0.20, 0.29) //from original matrix
= 0.29

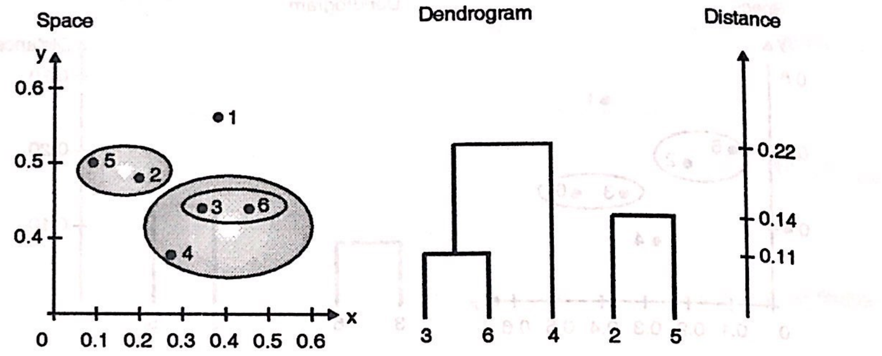
Therefore new distance matrix is :

| | | | | |
|---|---|---|---|---|
| p1 | 0 | | | |
| **(p2, p5)** | **0.34** | 0 | | |
| (p3, p6) | 0.23 | **0.39** | 0 | |
| p4 | 0.37 | **0.29** | 0.22 | 0 |
| | p1 | **(p2, p5)** | (p3, p6) | p4 |

**Step 5 :** Consider the following matrix

| | | | | |
|---|---|---|---|---|
| p1 | 0 | | | |
| **(p2, p5)** | **0.34** | 0 | | |
| (p3, p6) | 0.23 | **0.39** | 0 | |
| p4 | 0.37 | **0.29** | 0.22 | 0 |
| | p1 | **(p2, p5)** | (p3, p6) | p4 |

The minimum distance is 0.22, so merge (p3, p6) and p4




*dist*( (p3, p6, p4) , p1 ) = MAX ( *dist*(p3, p1) , *dist*(p6, p1), dist(p4, p1) )

= MAX (0.22, 0.23, 0.37) //from original matrix

= 0.37

*dist*( (p3, p6, p4) , (p2,p5) ) = MAX ( *dist*(p3, p2) , *dist*(p3, p5),

dist(p6, p2),dist(p6, p5),

dist(p4, p2) , dist(p4, p5) )

= MAX (0.15, 0.28, 0.25, 0.39, 0.20, 0.29)

= 0.39

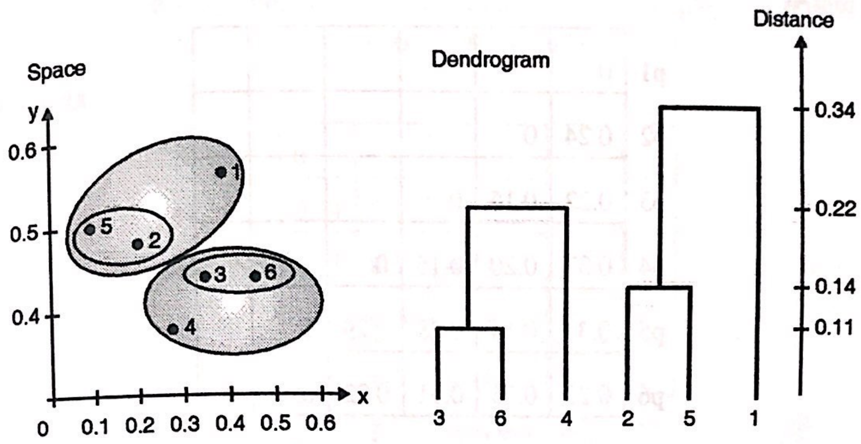
Therefore new distance matrix

| | p1 | **(p2, p5)** | (p3, p6, P4) |
|---|---|---|---|
| p1 | 0 | | |
| **(p2, p5)** | **0.34** | 0 | |
| (p3, p6, p4) | 0.37 | **0.39** | 0 |

**Step 6 :** Now consider the following distance matrix

| | p1 | **(p2, p5)** | (p3, p6, P4) |
|---|---|---|---|
| p1 | 0 | | |
| **(p2, p5)** | **0.34** | 0 | |
| (p3, p6, p4) | 0.37 | **0.39** | 0 |

Since the minimum distance is 0.34, merge (p2, p5) with p1.

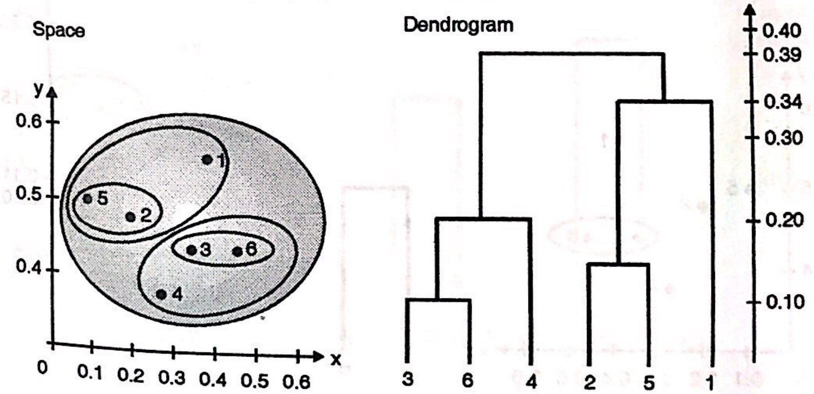



dist ((p2, p5, p1), (p3, p6, p4) = 0.39

Therefore new distance matrix

| | (p2, p5,p1) | (p3, p6, P4) |
|---|---|---|
| (p2, p5,p1) | 0 | |
| (p3, p6, p4) | 0.39 | 0 |

Finally, merge the cluster (p2, p5,p1) and (p3,p6,p4)




**Average link :**

**Step 1 and Step 2 :**

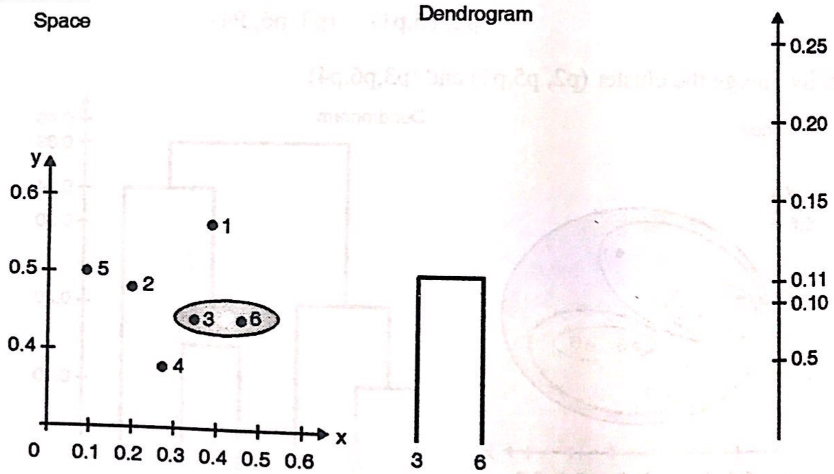
Refer the single link solution

**Distance matrix :**

| | p1 | p2 | p3 | p4 | p5 | p6 |
|---|---|---|---|---|---|---|
| p1 | 0 | | | | | |
| p2 | 0.24 | 0 | | | | |
| p3 | 0.22 | 0.15 | 0 | | | |
| p4 | 0.37 | 0.20 | 0.15 | 0 | | |
| p5 | 0.34 | 0.14 | 0.28 | 0.29 | 0 | |
| p6 | 0.23 | 0.25 | 0.11 | 0.22 | 0.39 | 0 |

**Step 3 :** Identify the two clusters with the shortest distance in the matrix, and merge them together. Re-compute the distance matrix, as those two clusters are now in a single cluster.

By looking at the distance matrix above, we see that p3 and p6 have the smallest distance from all – 0.11 So, we merge those two in a single cluster, and re-compute the distance matrix.




$$\text{dist}((p3\ p6),\ p1) = 1/2\ (\text{dist}(p3,p1) + \text{dist}(p6,p1))$$
$$= 0.5 \times (0.22 + 0.23) = 0.23$$
$$\text{dist}((p3\ p6),\ p2) = 1/2\ (\text{dist}(p3,p2) + \text{dist}(p6,p2))$$
$$= 0.5 \times (0.15 + 0.25) = 0.2$$
$$\text{dist}((p3\ p6),\ p4) = 1/2\ (\text{dist}(p3,p4) + \text{dist}(p6,p4))$$
$$= 0.5 \times (0.15 + 0.22) = 0.19$$

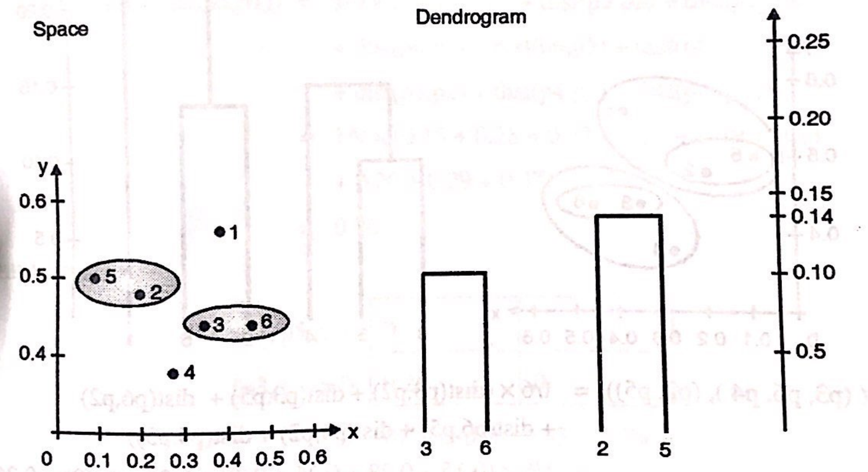
dist( (p3 p6), p5 ) = 1/2 (dist(p3,p5) + dist(p6,p5))
= $0.5 \times (0.28 + 0.39) = 0.34$

**Distance matrix :**

| | | | | | |
|---|---|---|---|---|---|
| p1 | 0 | | | | |
| p2 | 0.24 | 0 | | | |
| **(p3, p6)** | **0.23** | **0.2** | 0 | | |
| p4 | 0.37 | 0.20 | **0.19** | 0 | |
| p5 | 0.34 | 0.14 | **0.34** | 0.29 | 0 |
| | p1 | p2 | **(p3, p6)** | p4 | p5 |

**Step 4 :**

So, looking at the above distance matrix above, we see that p2 and p5 have the smallest distance from all – 0.14. So, we merge those two in a single cluster, and re-compute the distance matrix.




dist( (p2, p5), p1 ) = 1/2 (dist(p2,p1) + dist(p5,p1))
= $0.5 \times (0.24 + 0.34) = 0.29$

dist( (p2, p5), (p3,p6) ) = 1/4 (dist(p2,p3) + dist(p2,p6) + dist(p5,p3) + dist(p5,p6))
= $1/4 \times (0.15 + 0.25 + 0.28 + 0.39) = 0.27$

dist( (p2, p5), p4 ) = 1/2 (dist(p2,p4) + dist(p5,p4))
= $0.5 \times (0.14 + 0.29) = 0.22$

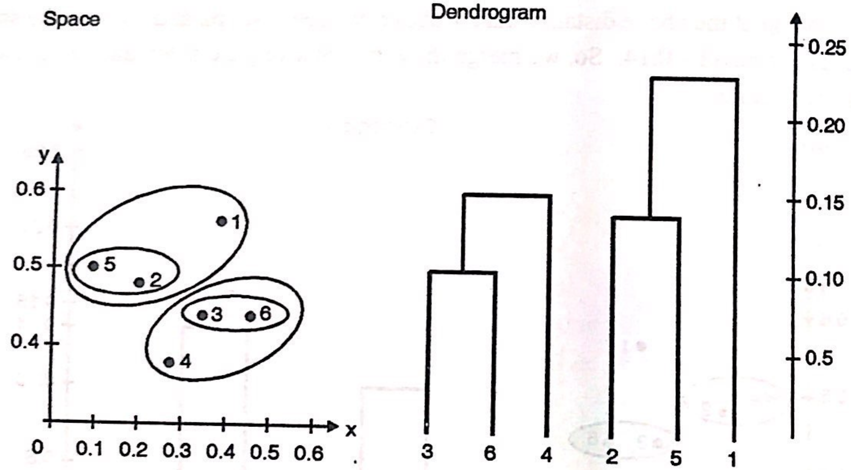
**Distance matrix :**

| | | | | |
|---|---|---|---|---|
| p1 | 0 | | | |
| **(p2, p5)** | **0.29** | 0 | | |
| (p3, p6) | 0.22 | **0.27** | 0 | |
| p4 | 0.37 | **0.22** | 0.15 | 0 |
| | p1 | **(p2, p5)** | (p3, p6) | p4 |

Since, we have merged (p2, p5) together in a cluster, we now have one entry for (p2, p5) in the table, and no longer have p2 or p5 separately.

**Step 5 :**

Now the closest clusters are merged, where distance is the smallest measure by looking at the maximum distance between any two points.




dist( (p3, p6, p4 ), (p2, p5)) = 1/6 × (dist(p3,p2) + dist(p3,p5) + dist(p6,p2) + dist(p6,p5) + dist(p4,p2) + dist(p4,p5))

= 1/6 × (0.15 + 0.28 + 0.25 + 0.39 + 0.20 + 0.29) = 0.26

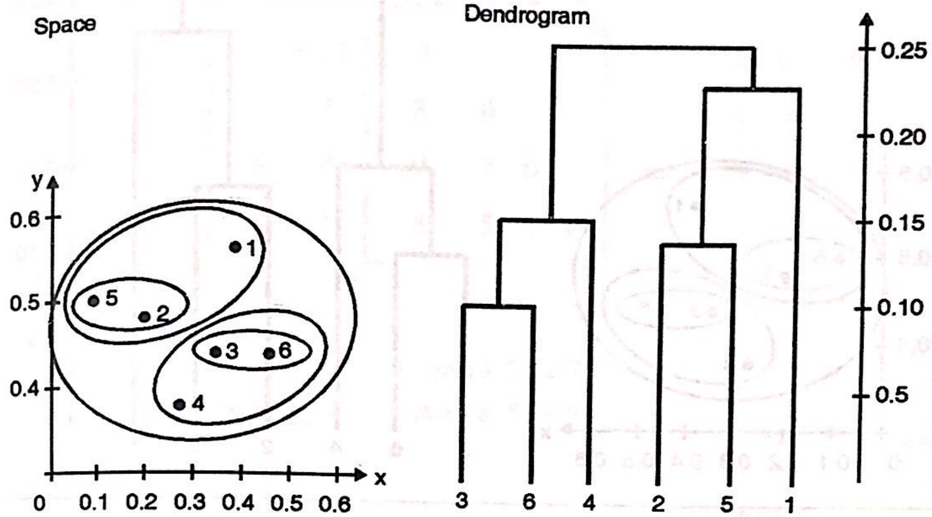
dist( (p3, p6, p4 ), p1) = 1/3 × (dist(p3,p1) + dist(p6,p1) + dist(p4,p1))

= 1/3 × (0.22 + 0.23 + 0.37) = 0.27

**Distance matrix :**

| | | | |
|---|---|---|---|
| p1 | 0 | | |
| **(p2, p5)** | **0.24** | 0 | |
| (p3, p6, p4) | 0.27 | **0.26** | 0 |
| | p1 | **(p2, p5)** | (p3, p6, P4) |

*Step 6 :* So merge the cluster (p2,p5) and p1.




$$
\begin{aligned}
dist((p3\ p6\ p4), (p2,p5,p1)) &= 1/9 \times (dist(p3,p2) + dist(p3,p5) + dist(p3,p1) \\
&\quad + dist(p6,p2) + dist(p6,p5) + dist(p6,p1) \\
&\quad + dist(p4,p2) + dist(p4,p5) + dist(p4+p1)) \\
&= 1/9 \times (0.15 + 0.28 + 0.22 + 0.25 + 0.39 + 0.23 \\
&\quad + 0.20 + 0.29 + 0.37) \\
&= 0.26
\end{aligned}
$$

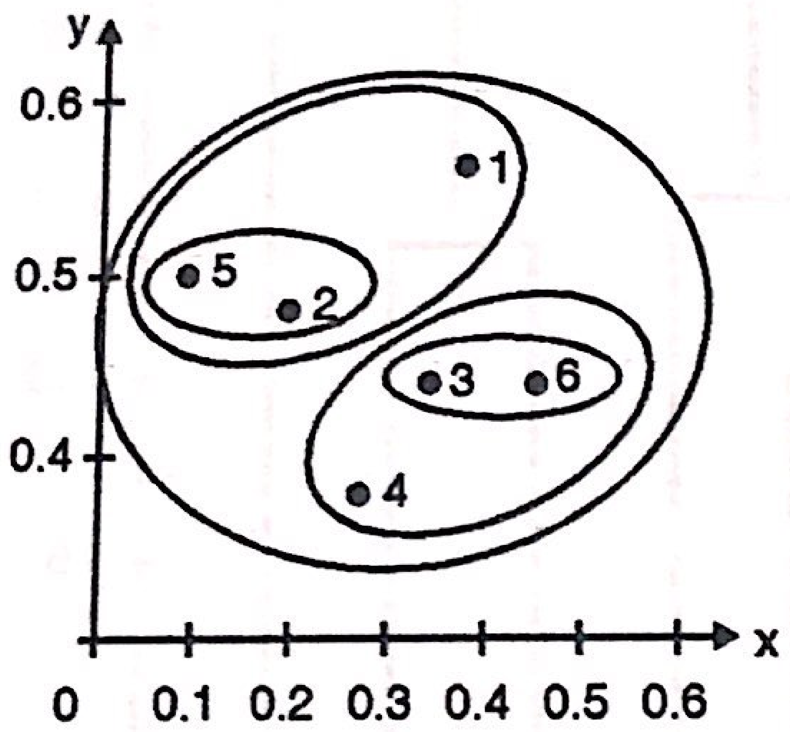
**Distance matrix :**

| | (p2, p5,p1) | (p3, p6, P4) |
|---|---|---|
| (p2, p5,p1) | 0 | |
| (p3, p6, p4) | 0.26 | 0 |

We need to re-compute the distance from all other points / clusters to our new cluster - (p2, p5,p1) to (p3,p6,p4) and enter the maximum distance in the above matrix (use original distance matrix).

Finally merge the cluster (p2, p5,p1) and (p3,p6,p4).

Space




Dendrogram